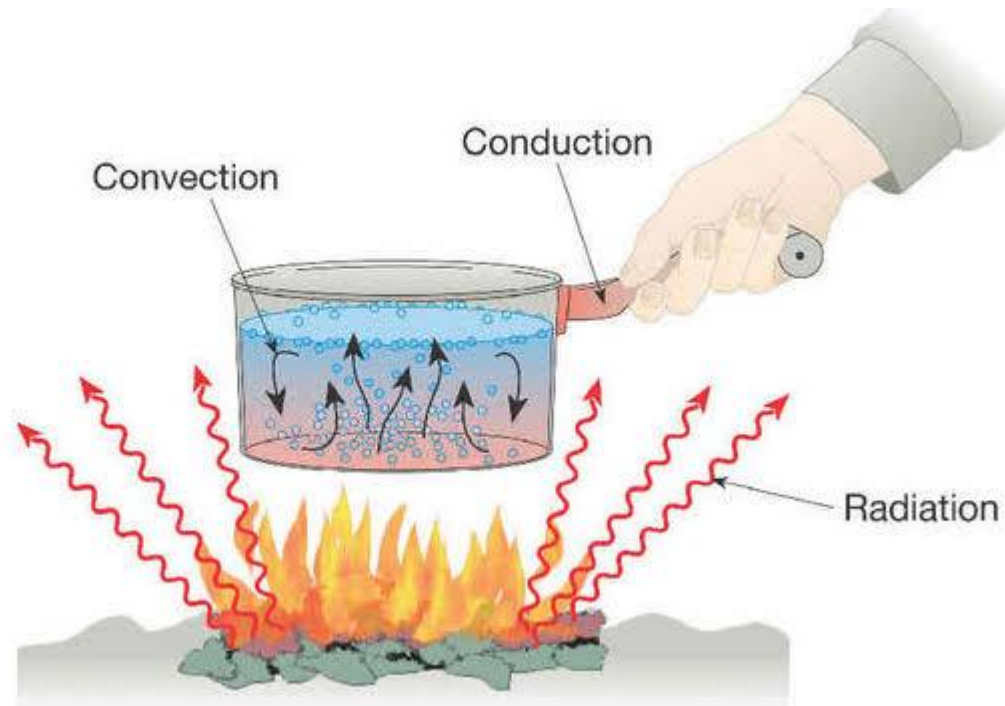


HEAT & THERMODYNAMICS

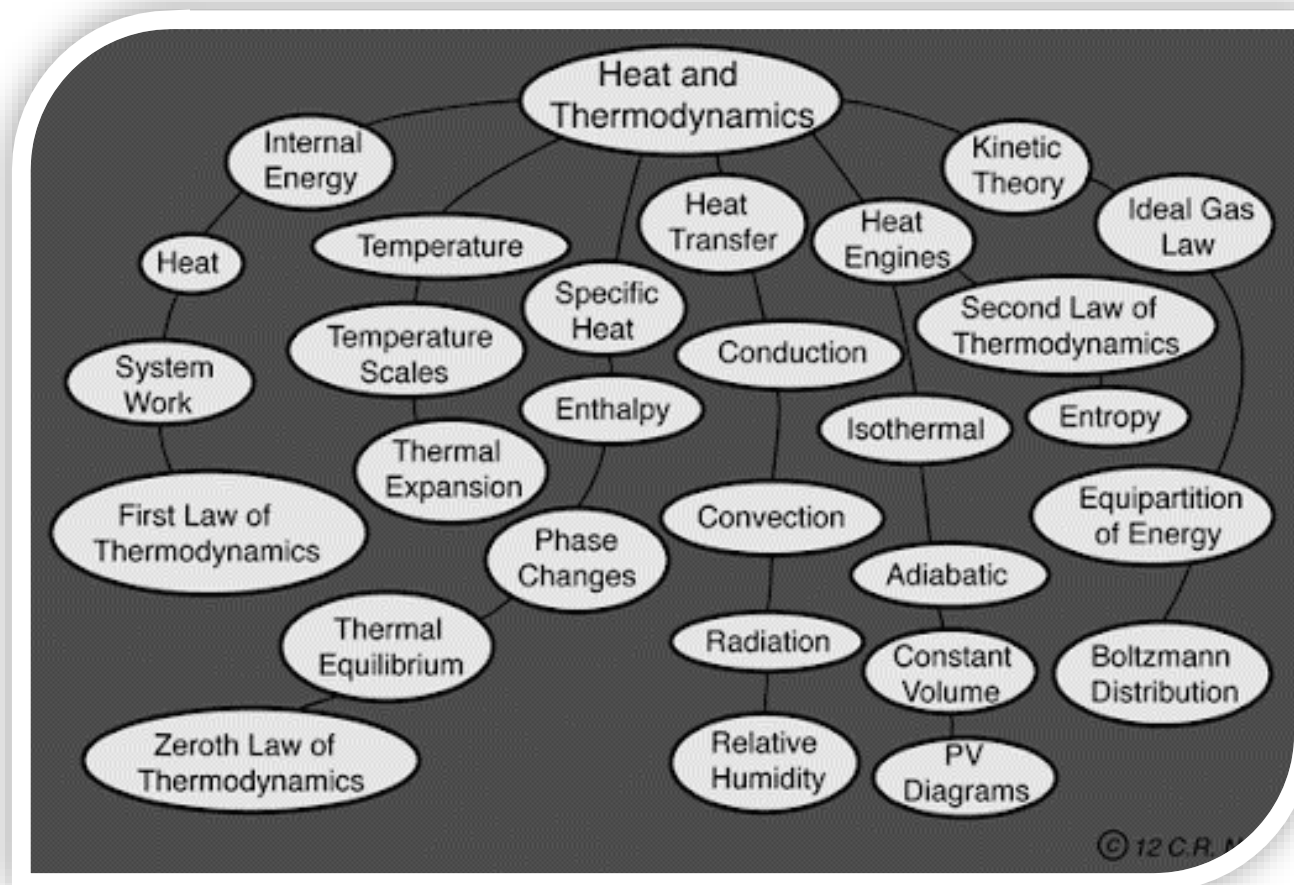


Content

- Platinum resistance thermometer,
- Thermo-electric thermometer,
- Kinetic theory of gases,
- Maxwell's distribution of molecular speeds,
- Mean free path,
- Equipartition of energy,
- Brownian motion,
- Vander Waal's equation of state,
- First Law of thermodynamics and its application,

- Reversible and irreversible processes,
- Second Law of thermodynamics,
- Carnot cycle,
- Efficiency of heat engines,
- Carnot's Theorem,
- Entropy and disorder,
- Thermodynamic functions,
- Maxwell relations,
- Third Law of thermodynamics.

Thermodynamics: The branch of physical science that deals with the relations between heat and other forms of energy (such as mechanical, electrical or chemical energy) and the relationships between all forms of energy.



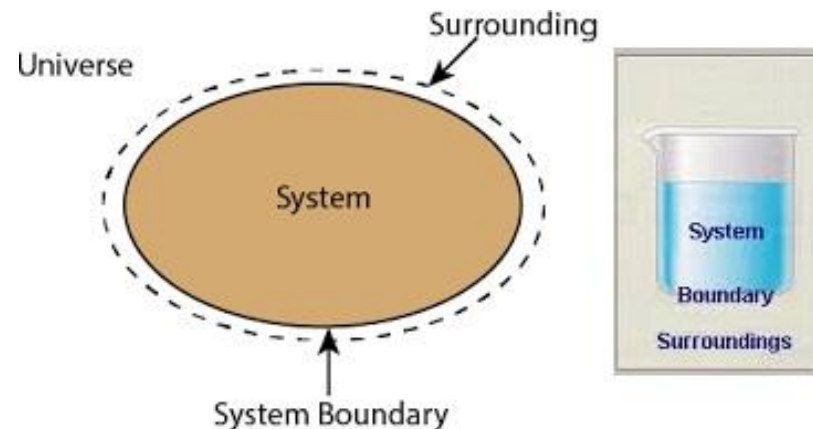
Differences between Heat and Temperature

Heat	Temperature
Heat is a type of energy.	Temperature is not energy. It is a thermal condition of body.
Its SI unit joule.	Its SI unit kelvin.
Heat depends on mass of the substance. One glass of boiling water and one teapot of boiling tea have different heat since they have different masses.	Temperature does not depend on the mass of matter. For example, temperature of one glass of boiling water and one teapot of boiling tea are equal to each other.
Heat cannot be measured with a device directly. You should know the mass, temperature and specific heat capacity of that matter.	Temperature can measure directly with a device called thermometer.
It is an cause.	It is an effect.

Thermodynamic System

A thermodynamic system is a quantity of matter of fixed identity, around which we can draw a boundary. The boundaries may be fixed or moveable. Work or heat can be transferred across the system or boundary.

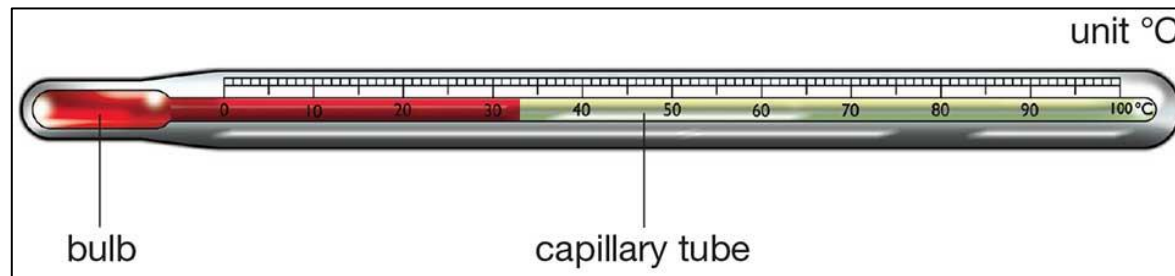
Examples: Refrigerator, air-conditioner, washing machine, heat exchange, a utensil with hot water etc.



Open system	Closed system	Isolated system
The system in which the transfer of mass as well as energy can take place across its boundary is called as an open system.	The system in which the transfer of energy takes place across its boundary with the surrounding, but no transfer of mass takes place is called as closed system. The closed system is fixed mass system.	The system in which neither the transfer of mass nor that of energy takes place across its boundary with the surroundings is called as isolated system. Here there will neither transfer of mass nor that of energy.
Example: Open system is boiling water in an open vessel, where transfer of heat as well as mass in the form of steam takes place between the vessel and surrounding.	Example: The water being heated in the closed vessel, when water will get heated but its mass will remain same.	Hot water, coffee or tea kept in the thermos flask is closed system.
		

Thermometer

An instrument for measuring and indicating temperature. Typically, it consists of a narrow, hermetically sealed glass tube marked with graduations and having at one end a bulb containing mercury or alcohol that expands and contracts in the tube with heating and cooling.



Some Types of the thermometer

Mercury thermometers

Alcohol in glass thermometers

Constant pressure gas thermometers

Constant volume thermometers

Platinum resistance thermometers

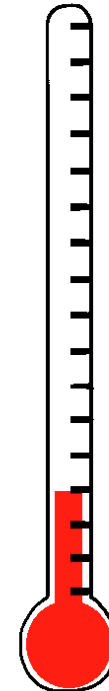
Thermocouple thermometers

Pyrometer Thermometers

Why mercury used mostly in thermometers instead of water?

Mercury is used mostly in thermometers because of the following properties:

- It is visible.
- It has a low freezing point ($-39\text{ }^{\circ}\text{C}$).
- It has a very high boiling point ($357\text{ }^{\circ}\text{C}$).
- It expands linearly.
- It gives accurate measurements.
- It is a good conductor of heat.
- Fast response time.
- It has a wide range of temperatures.



Platinum resistance thermometer

The Platinum resistance thermometer is based on the principle that electrical resistance of a metallic wire increase gradually and uniformly with increase in temperature over a wide range. The scientist Clausius was first who develop a relation as,

$$R_{\theta} = R_0(1 + \alpha\theta)..... (1)$$

Where, R_0 and R_{θ} are the resistances of the conducting wire at $0^{\circ}C$ and $\theta^{\circ}C$ respectively and α be the coefficient of increase in resistance whose value depend on the nature and property of the wire.

If R_0 , R_{100} and R_θ be the resistances of a pure platinum wire at 0°C , 100°C and unknown temperature $\theta_p^\circ\text{C}$ respectively, then by using equation (1) we can write,

$$R_{100} = R_0(1 + \alpha \cdot 100)$$

$$\therefore \alpha = \frac{R_{100} - R_0}{100R_0} \dots\dots\dots (2)$$



$$\text{And, } R_{\theta} = R_0(1 + \alpha \cdot \theta_p)$$

$$\theta_p = \frac{R_{\theta} - R_0}{\alpha R_0} = \frac{R_{\theta} - R_0}{\left(\frac{R_{100} - R_0}{100 R_0}\right) \cdot R_0}$$

$$= \frac{R_{\theta} - R_0}{R_0} \times \frac{100 R_0}{R_{100} - R_0}$$

$$\therefore \theta_p = \frac{R_{\theta} - R_0}{R_{100} - R_0} \times 100 \dots\dots\dots (3)$$

By knowing the values of R_0 , R_{100} and R_{θ} we can determine the value of $\theta_p^0 C$ from equation (3).

1. The resistance of the platinum wire of a platinum resistance thermometer at the ice point is $5\ \Omega$ and at the steam point is $5.93\ \Omega$. When the thermometer is inserted in a hot bath the resistance of the platinum wire is $5.795\ \Omega$. Calculate the Celsius temperature on the platinum scale. [Ans: $85.48\ ^\circ\text{C}$]
2. The values of resistance of a platinum resistance thermometer are $2.585\ \text{ohms}$ and $3.510\ \text{ohms}$ at 0°C and 100°C respectively. When placed in a hot bath, the resistance is found to be $9.098\ \text{ohms}$. Calculate the temperature of the hot bath on the platinum scale.
3. The resistance of a platinum resistance thermometer are $2.56\ \text{ohms}$ and $3.56\ \text{ohms}$ at 0°C and at steam point. When the thermometer is placed in an unknown temperature bath, the resistance of the thermometer is $6.78\ \text{ohms}$. Calculate the temperature of the bath?
4. What is the resistance of a platinum wire at 25°C ? Here, resistance at 0°C , $R_0 = 2.585\ \Omega$, $\alpha = 3.8 \times 10^{-3}$
5. A platinum wire has resistances of $2\ \Omega$ and $3\ \Omega$ respectively at 0°C and 100°C . Its resistance becomes $10.5\ \Omega$ at the normal boiling point of sulphur. Calculate the value of boiling point of sulphur from the given data.

FOUR MAJOR TEMPERATURE SCALES

The **Fahrenheit scale** of temperature is the common form of temperature measurement used in the United States and some parts of the Caribbean. It was created by the German scientist **Daniel Gabriel Fahrenheit** in the early 18th century, and adapted its measurements standards from a previous scale created by Ole Roemer. Water freezes at 32 degrees Fahrenheit, and boils at 212 degrees F. The Fahrenheit temperature scale includes negative temperatures, below 0 degrees F. The coldest possible temperature, absolute zero, is -459.67 degrees F.

Two versions of the **Celsius scale** were created in the early 18th century – one by Swedish scientist **Anders Celsius**, and another by the French Jean Pierre Cristin. The Celsius scale is sometimes referred to as the centigrade scale, because it is based on a 100 degree division between the freezing and boiling points of water. Water freezes at 0 degrees Celsius and boils at 100 degrees C.

The **Kelvin scale** was adapted from the Celsius scale in the 19th century by the British scientist William Thompson, later **Lord Kelvin**. Kelvin was designed in order to set the zero point of the temperature scale at absolute zero. Because of this, absolute zero is located at 0 K – Kelvin does not use degrees in its notation. You can convert from Celsius to Kelvin by adding 273.15 to a Celsius temperature. Water freezes at 273.15 K, and boils at 373.15 K .

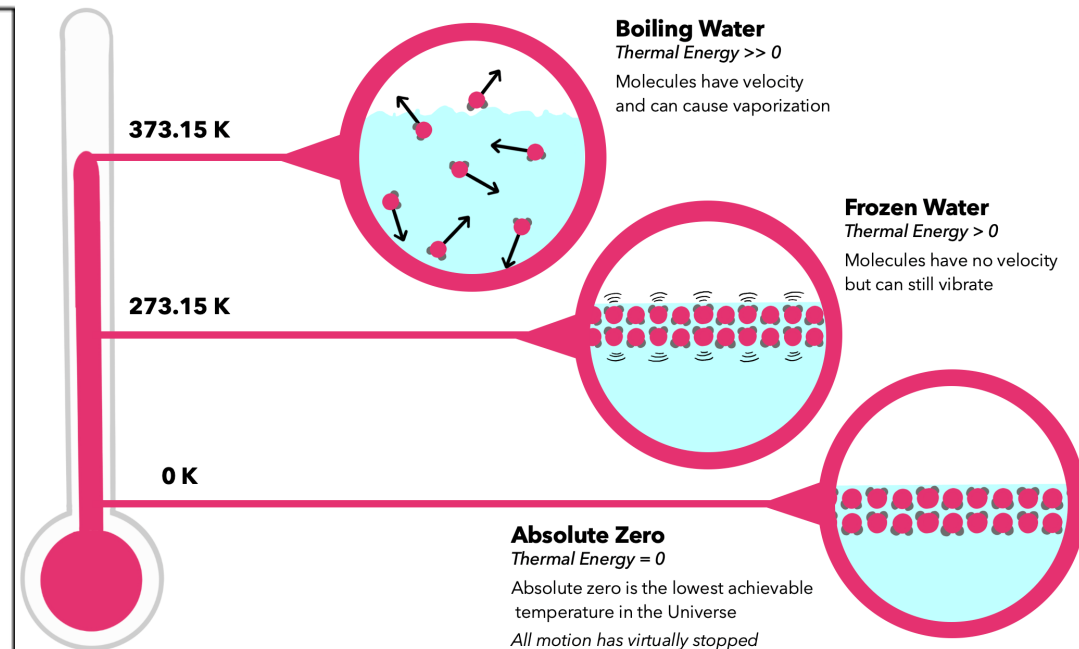
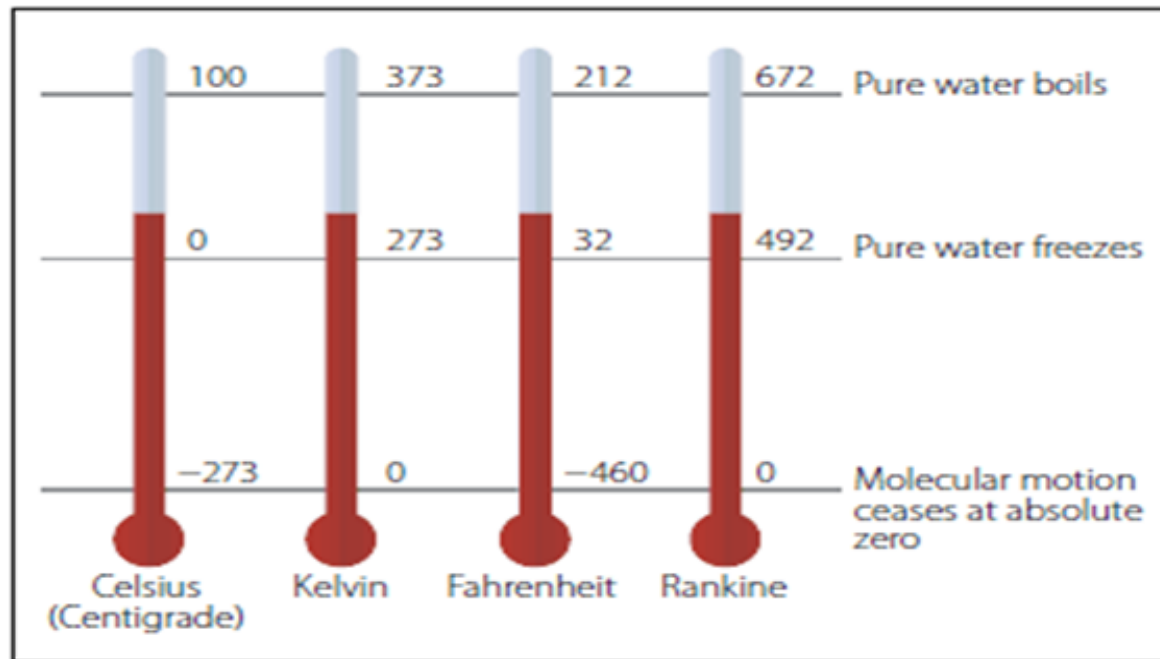
While not widely used – apart from some U.S. engineering fields – the **Rankine scale** provides an absolute zero-based equivalent to the Fahrenheit scale. The scale was created by Scottish scientist **William John Rankine** in the 19th century, shortly after the creation of the Kelvin scale. Temperatures can be converted from Fahrenheit to Rankine by adding 459.67. Absolute zero is thus located at 0 degrees Rankine. Water freezes at 491.67 degrees R, and boils at 671.67 degrees R.

Temperature Scale

$$\frac{C - 0}{100 - 0} = \frac{F - 32}{212 - 32} = \frac{K - 273}{373 - 273} = \frac{R - 492}{672 - 492}$$

$$\frac{C}{100} = \frac{F - 32}{180} = \frac{K - 273}{100} = \frac{R - 492}{180}$$

$$\frac{C}{5} = \frac{F - 32}{9} = \frac{K - 273}{5} = \frac{R - 492}{9}$$

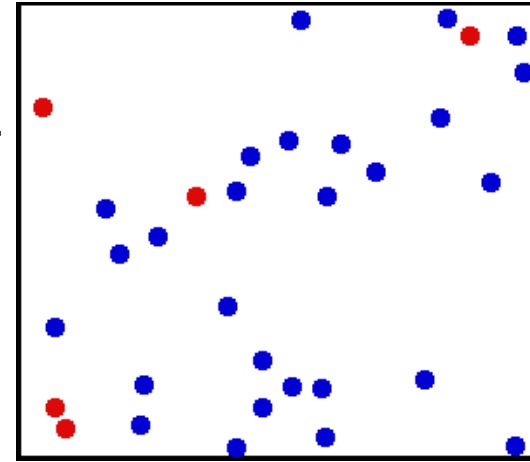


1. The temperature of the surface of the sun is about 6500°C . What is the temperature a) On the Rankin scale? b) On the Kelvin scale.
2. The normal boiling point of liquid oxygen is -183°C . What is this temperature on i) Kelvin scale and ii) Rankine scale?
3. One day when temperature is 15°F . What will be the temperature on Celsius and Kelvin scale?
4. At what temperature do the Kelvin and Fahrenheit scales coincide?
5. At what temperature do the Celsius and Fahrenheit scales coincide?
6. Which temperature has 40° differences in Celsius and Fahrenheit scale?

KINETIC THEORY OF GASES

The **kinetic theory of gases** is a simple, historically significant model of the [thermodynamic](#) behavior of [gases](#), with which many principal concepts of thermodynamics were established. The model describes a gas as a large number of identical submicroscopic [particles](#) ([atoms](#) or [molecules](#)), all of which are in constant, rapid, [random motion](#). Their size is assumed to be much smaller than the average distance between the particles. Kinetic Theory of Gases is based on the following assumption, first stated by Clausius in 1860.

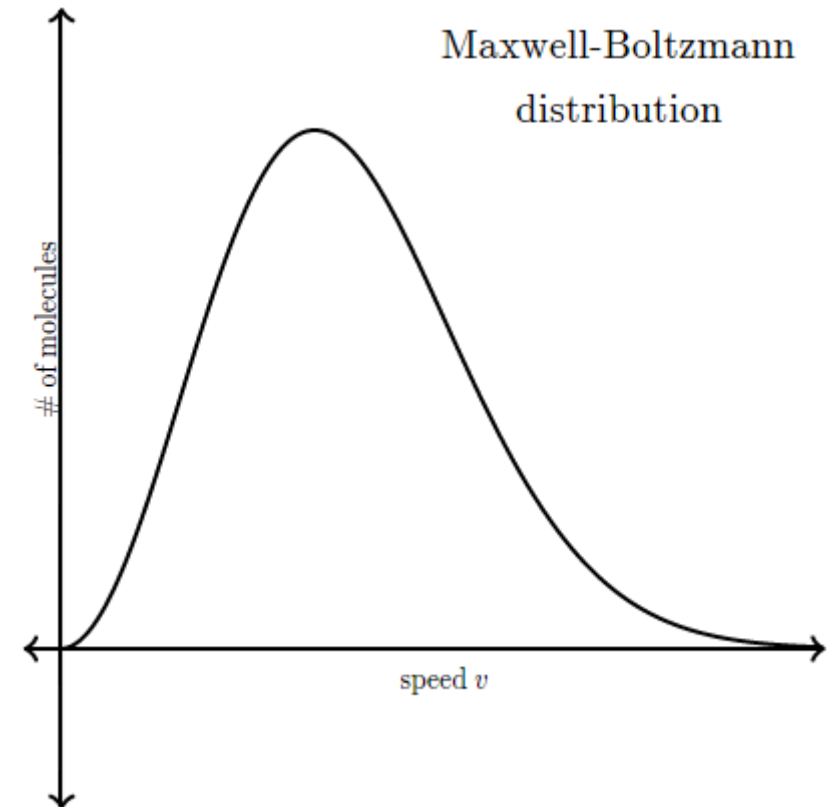
1. A gas consists of large number of tiny, rigid particles called molecules.
2. The gas molecules are identical in all respects, like mass, size, etc.
3. The volume of the molecules is negligible when compared to the volume of the gas.
4. Molecules are perfectly elastic spheres and do not exert forces on each other.
5. The molecules are in random motion and obey Newton's law.



Maxwell's Distribution of Molecular Speeds

In the context of the Kinetic Molecular Theory of Gases, a gas contains a large number of particles in rapid motions. Each particle has a different speed, and each collision between particles changes the speeds of the particles. An understanding of the properties of the gas requires an understanding of the distribution of particle speeds.

If we were to plot the number of molecules whose velocities fall within a series of narrow ranges, we would obtain a slightly asymmetric curve known as a velocity distribution. This velocity distribution curve is known as the Maxwell-Boltzmann distribution. The **Maxwell-Boltzmann distribution law** was first worked out around 1850 by the great Scottish physicist, **James Clerk Maxwell** and Later, the Austrian physicist **Ludwig Boltzmann**. Notice that the graph is not symmetrical. There is a longer "tail" on the high speed right end of the graph. The graph continues to the right to extremely large speeds, but to the left the graph must end at zero (since a molecule can't have a speed less than zero)



The Maxwell-Boltzmann equation, which forms the basis of the kinetic theory of gases, defines the distribution of speeds for a gas at a certain temperature. The Maxwell-Boltzmann distribution is used to determine how many molecules are moving between velocities v and $v+dv$. The Maxwell-Boltzmann distribution is given by,

$$dn = 4\pi n \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-\frac{mv^2}{2k_B T}} dv, \dots\dots\dots (1)$$

n = density of the gas,

k_B = Boltzmann constant,

T = temperature of the gas,

V = velocity of the gas molecule

The total number of molecules in the gas can be obtained by,

$$n = \int_0^\infty 4\pi n \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-\frac{mv^2}{2k_B T}} dv \dots\dots\dots (2)$$

Three speed expressions can be derived from the Maxwell-Boltzmann distribution:

- **Most probable speed:** The most probable speed is the speed associated with the highest point in the Maxwell distribution. Only a small fraction of particles might have this speed, but it is more likely than any other speed.

$$v_{mp} = \sqrt{\frac{2RT}{M}}$$

- **Average speed:** The average speed is the sum of the speeds of all of the particles divided by the number of particles

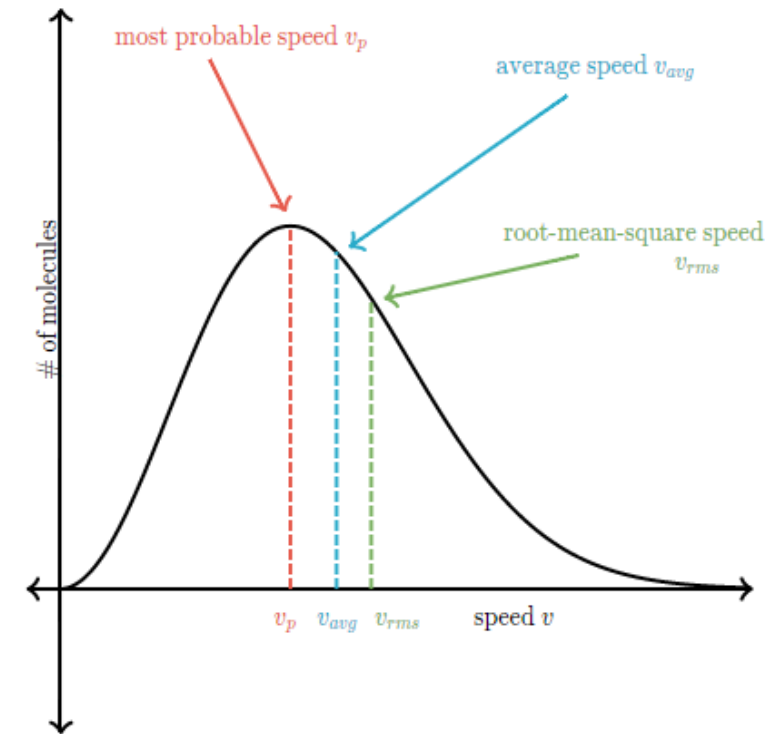
$$v_{avg} = \sqrt{\frac{8RT}{\pi M}}$$

- **Root-mean-square speed:** The root-mean-square speed is square root of the average speed-squared.

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

Where, M=Molar mass which is the mass of one mole of a substance.
Molar mass is reported in grams per mole or g/mol

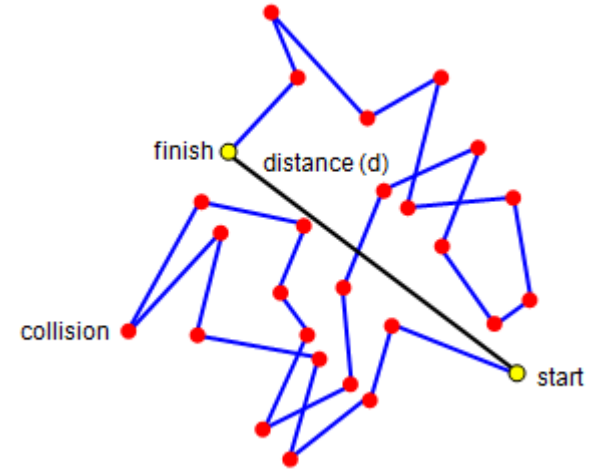
A plot of n against v is given below--



Mean Free Path

The average distance travelled by a molecule between two collisions is called the mean free path (λ). If S is the total distance travelled during N collisions, then

$$\lambda = \frac{S}{N}$$



Degrees of Freedom

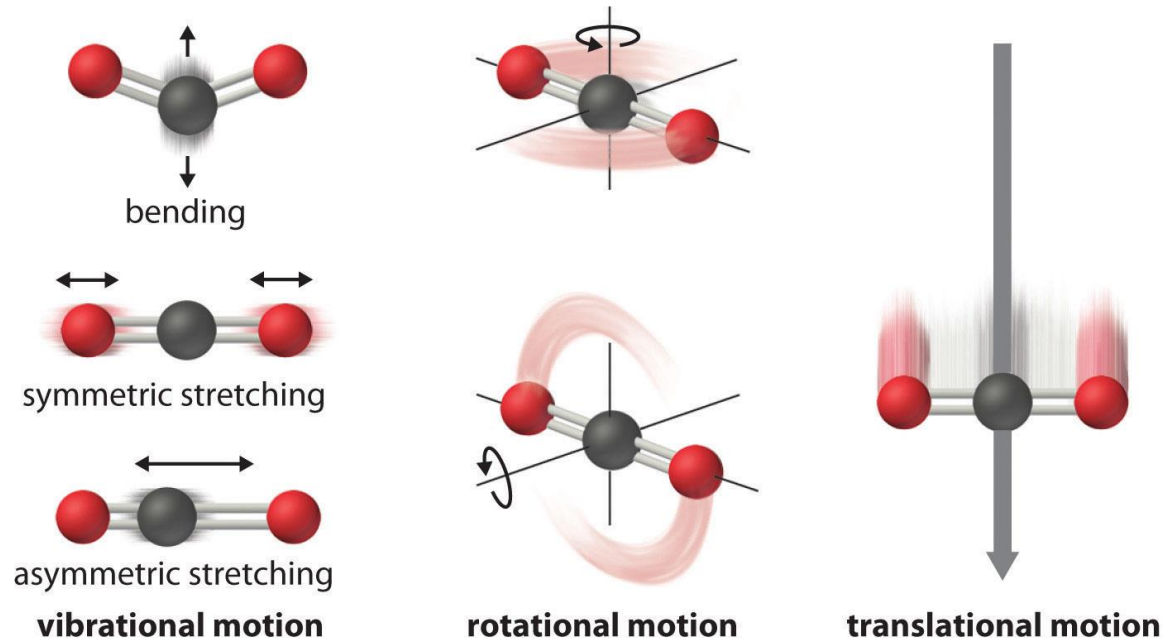
“The total number of independent variables required to describe completely the state of motion of a body are called its degrees of freedom”.

An ant moving along a straight line, it has only **one** degree of freedom (x).

If it moves on a plane it has **two** degrees of freedom (x,y).

A flying mosquito has **three** degrees of freedom (x,y,z). Also a monoatomic molecule, e.g., Neon (Ne), Helium (He) etc., has three degrees of freedom (all are translational)

A diatomic molecule having a dumbbell shape e.g., HCl, Cl₂, O₂ etc., has **six** degrees of freedom (3 translational, 2 rotational and 1 vibrational), as shown in figure.



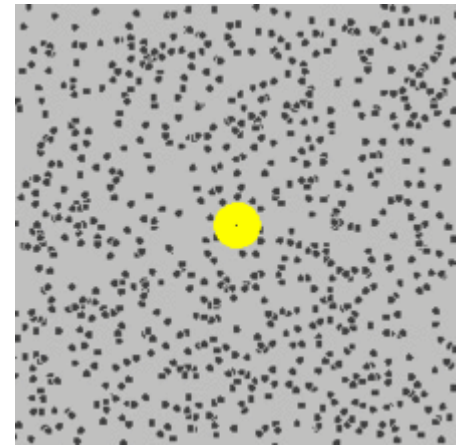
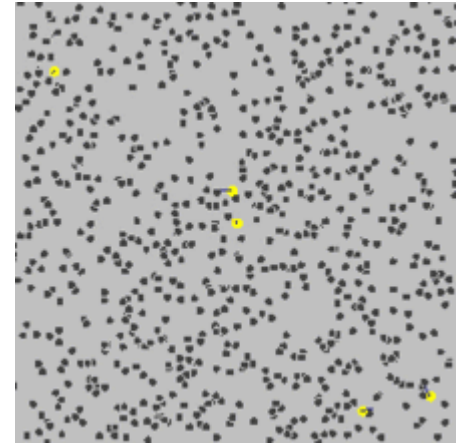
Principle of Equipartition of Energy

Equipartition of energy is an equal amount of energy will be associated with each [degree of freedom](#). The average kinetic energy associated with each degree of freedom is $\frac{1}{2} kT$

Statement: “For any dynamical system in thermal equilibrium, the total energy is divided equally among all the degrees of freedom and the energy associated per molecule per degree of freedom is $\frac{1}{2} kT$, where k is Boltzmann’s constant and T is absolute temperature of the gas”.

Brownian Motion

- Particles in both liquids and gases (collectively called fluids) move randomly. This is called Brownian motion. They do this because they are bombarded by the other moving particles in the fluid. Larger particles can be moved by light, fast-moving molecules.
- Brownian motion is named after the **botanist Robert Brown**, who first observed this in 1827. He used a microscope to look at pollen grains moving randomly in water. At this point, he could not explain why this occurred. But in 1905, physicist **Albert Einstein** explained that the pollen grains were being moved by individual water molecules. This confirmed that atoms and molecules did exist, and provided evidence for **particle theory**.



Vander Waal's Equation of State

In thermodynamics, the Van der Waals equation (or Van der Waals equation of state) named after Dutch physicist Johannes Diderik van der Waals, is an equation of state that generalizes the ideal gas law based on plausible reasons that real gases do not act ideally. The ideal gas law treats gas molecules as point particles that interact with their containers but not each other, meaning they neither take up space nor change kinetic energy during collisions. The ideal gas law states that volume (V) occupied by n moles of any gas has a pressure (P) at temperature (T) in kelvins given by the following relationship, where R is the gas constant:

$$PV = nRT$$

To account for the volume that a real gas molecule takes up, the Van der Waals equation replaces V in the ideal gas law with $V_m - b$, where V_m is the molar volume of the gas and b is the volume that is occupied by one mole of the molecules. This leads to

$$P(V_m - b) = RT$$

The second modification made to the ideal gas law accounts for the fact that gas molecules do in fact interact with each other (they usually experience attraction at low pressures and repulsion at high pressures) and that real gases therefore show different compressibility than ideal gases. Van der Waals provided for intermolecular interaction by adding to the observed pressure P in the equation of state a term $\frac{a}{V_m^2}$ where a is a constant whose value depends on the gas. The Van der Waals equation is therefore written as:

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

and, for n moles of gas, can also be written as the equation below

$$\left(P + \frac{an^2}{V^2}\right)(V_m - nb) = nRT$$

Zeroth Law of Thermodynamics

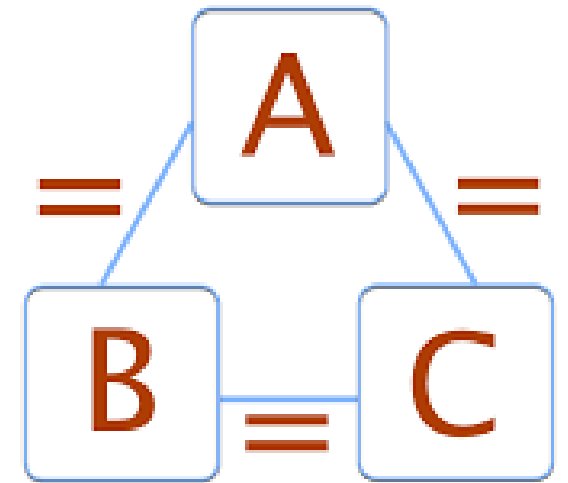
- Zeroth law of thermodynamics is one of the four laws of thermodynamics. The zeroth law of thermodynamics frames an idea of temperature as an indicator of thermal equilibrium.
- **Thermal Equilibrium:** Temperature is a property that distinguishes thermodynamics from other sciences. This property can distinguish between hot and cold. When two or more bodies at different temperatures are brought into contact then after some time they attain a common temperature and they are said to exist in thermal equilibrium. Systems are said to be in thermal equilibrium if there is no heat transfer, even if they are in a position to transfer heat, based on other factors. For example, if we put food in the refrigerator overnight then that food is in thermal equilibrium with the air of that refrigerator. Heat no longer flows from food to the air or from the air to the food, this state is known as thermal equilibrium.

Zeroth Law of Thermodynamics

Statement:

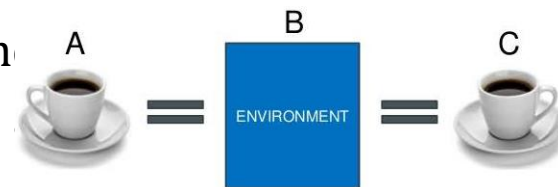
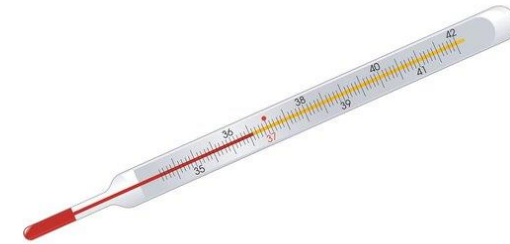
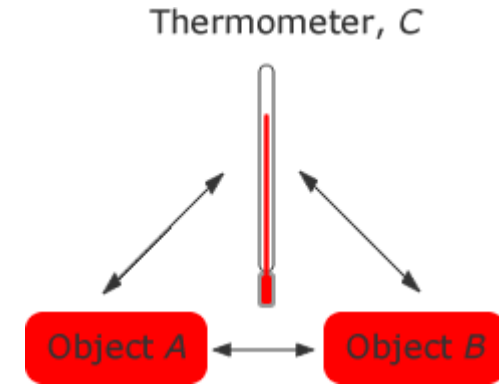
“The zeroth law of thermodynamics states that if two thermodynamic systems are each in thermal equilibrium with a third, then the two systems are in thermal equilibrium with each other also”.

When a body ‘A’ is in thermal equilibrium with another body ‘b’, and also separately in thermal equilibrium with a body ‘C’, then body ‘B’ and ‘C’ will also be in thermal equilibrium with each other. There are also various ways to state the zeroth law of thermodynamics. However, in simple terms, it can be said, “Systems that are in thermal equilibrium exist at the same temperature”.



Application of the Zeroth Law

- Let's say Object A is in "India" and Object B is in "Bangladesh" and we want some Information about the temperature of both. What will you do ? Here comes the Body C (Assume it to be mercury Thermometer). In both these country Body C is available. We measured the Height of Mercury Column in India after contacting it with Object A and also measured the same in Bangladesh after Contacting it with Object B. At both these places the Height of Mercury Comes out to be same. Therefore without bringing in contact physically the Object A and Object B we can easily conclude that they are at a same temperature or in a thermal Equilibrium.
- Measuring our body temperature using mercury thermometer:** When mercury thermometer is placed on our body, the glass wall of the thermometer first attains equilibrium with our body due to heat conduction. Then equilibrium is attained between the glass wall and the mercury. So if our body and mercury are in thermal equilibrium with the glass wall. then our body and mercury are in thermal equilibrium with each other ie.,mercury shows the exact body temperature.
- Another example of the zeroth law of thermodynamics is when you have two cup of coffee. On cup will have hot coffee and the other have cold coffee. Now if we leave them in the table for few hours they will attain thermal equilibrium with the temperature of the room.



First Law of Thermodynamics

- Before we get into the first law of thermodynamics we need to understand the relation between heat and work and the concept of internal energy. Just like mass, energy is always conserved i.e. it can neither be created nor destroyed but it can be transformed from one form to another.
- **Internal energy** is a thermodynamic property of the system that refers to the energy associated with the molecules of the system which includes kinetic energy (due to translational, rotational and vibrational motion of molecules which only depends on T) and potential energy (due to intermolecular force) which depends on the separation between the molecules.

First Law of thermodynamics

Statement: “ If some heat is supplied to a system which can do work, then the quantity of heat absorbed by the system is equal to the sum of increase in internal energy of the system and the external work done by the system”.

Explanation: If dQ be the amount of heat absorbed by a system and dU be the change in internal energy and dW is the work done by the system due to absorption of heat, then according to the 1st law of thermodynamics,

$$dQ = dU + dW$$

Limitations of First Law of thermodynamics

The first law states that whenever a system undergoes any thermodynamic process it always holds certain energy balance. But it fails to give the feasibility of the process or change of state that the system undergoes. For instance, the first law fails to explain why heat flows from hot end to cold end when a metallic rod is heated at one end and not on the other and vice-versa. The first law only quantifies the energy transfer that takes place during this process. It is the second law of thermodynamics which provides the criterion for the feasibility of the various processes.

Work Done in the Thermodynamic System

The work, dW done by the gas is given by,

$$dW = F dx \cos \theta$$

Where, $\theta = 0^\circ$ and $F = PA$

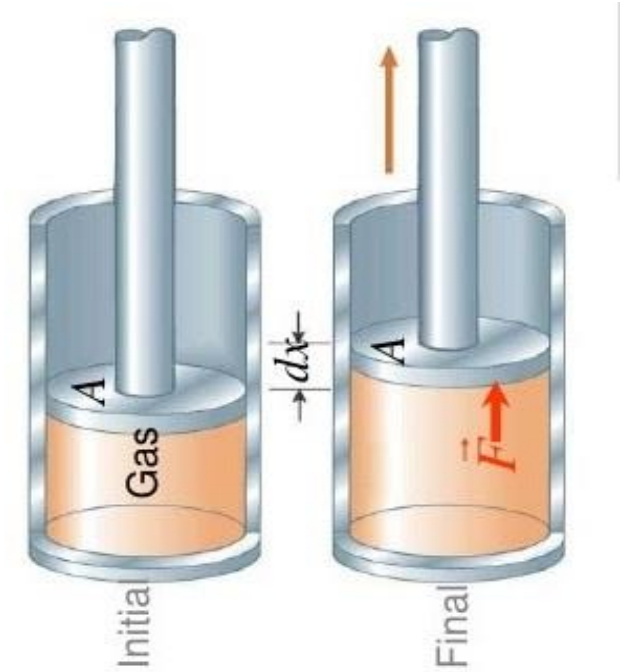
$$dW = PA dx$$

And $A dx = dV$

$$\therefore dW = PdV$$

In a finite change of volume from V_1 to V_2 ,

$$W = \int dW = \int_{V_1}^{V_2} PdV$$



Problem: Some gas are confined in a cylinder having movable piston. The system has done 1200 J work when 800 J heat energy is given keeping pressure constant at 400 Pa. What are the change in volume and internal energy?

Specific Heat (S): The specific heat of material is defined as the quantity of heat required to raise the temperature of unit 1 gm of that material through 1 degree Celsius.

$$\text{Specific heat, } S = \frac{Q}{m\Delta T}$$

The unit of specific heat is in C.G.S. is $\text{cal g}^{-1}\text{°C}^{-1}$ and in S.I. is $\text{JK g}^{-1}\text{K}^{-1}$.

Molar Specific Heat at Constant Volume (C_v): It is defined as the amount of heat energy required to raise the temperature of one mole of a gas by 1°C at **constant volume**. It is represented by C_v and given by,

$$C_v = \left(\frac{dQ}{n dT} \right)_v$$

Molar Specific Heat at Constant Pressure (C_p): It is defined as the amount of heat energy required to raise the temperature of one mole of a gas by 1°C at **constant pressure**. It is represented by C_p and given by,

$$C_p = \left(\frac{dQ}{n dT} \right)_p$$

Relation between C_p and C_v

Let us consider one mole of an ideal gas enclosed in a cylinder provided with a frictionless piston of area A . Let P , V and T be the pressure, volume and absolute temperature of gas respectively as shown in below figure.

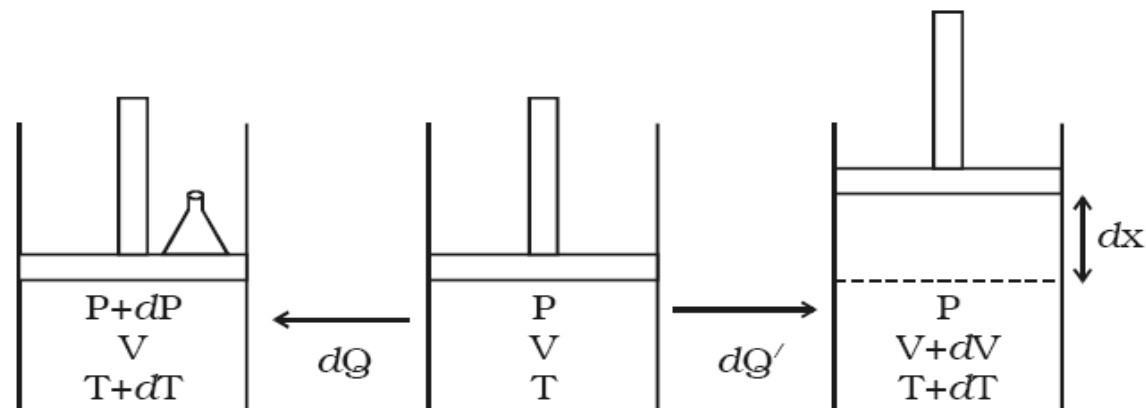
A quantity of heat dQ is supplied to the gas. To keep the volume of the gas constant, a small weight is placed over the piston. The pressure and the temperature of the gas increase to $P + dP$ and $T + dT$ respectively. This heat energy dQ is used to increase the internal energy dU of the gas

$$\therefore dQ = n \times C_v \times dT \quad \dots\dots\dots (1)$$

where, C_v is the molar specific heat at constant volume. According to first law of thermodynamics

$$dQ = dU + dW$$

$$\therefore dQ = dU + PdV \quad \dots\dots\dots (2)$$



Since the volume is constant, $dV = 0$.

From (2) we get,

$$dQ = dU \dots\dots\dots (3)$$

From (1) and (3) we have,

$$dU = nC_v dT \dots\dots\dots (4)$$

Now, keeping the pressure constant, the temperature of the system is raised by the same amount dT by supplying heat dQ .

So, we get the amount of heat,

$$dQ = nC_p dT \dots\dots\dots (5)$$

Using equations (4) and (5) in (2) we get,

$$nC_p dT = nC_v dT + PdV \dots\dots\dots (6)$$

But, for n moles of ideal gas,

$$PV = nRT \dots\dots\dots (7)$$

Differentiating (7) we get,

$$PdV = nRdT \dots\dots\dots (8)$$

Using (8) in (6) we get,

$$nC_p dT = nC_v dT + nRdT$$

$$\text{or, } nC_p = nC_v + nR$$

$$\text{or, } n(C_p - C_v) = nR$$

$$\text{or, } C_p - C_v = R \dots\dots\dots (9)$$

Therefore, the difference between the two specific heats of an ideal gas is equal to the molar gas constant, R . Now, from equation (9) we can write,

$$C_p = R + C_v$$

$\therefore C_p > C_v$ For an ideal gas the ratio between C_p and C_v is denoted by γ .

$$\therefore \gamma = \frac{C_p}{C_v}.$$

From experiment we have seen that, γ is 1.67 for one atomic gas, $\gamma = 1.41$ for two atomic gas and $\gamma = 1.33$ for three atomic gas.

Problem: For an ideal gas $\gamma = 1.4$. Calculate the values of molecular specific heats of the gas?

Solution:

We have, $C_p - C_v = R \dots \dots \dots (1)$

And $\gamma = C_p/C_v$

$$\text{or, } C_p = \gamma C_v$$

Substituting this in (1)

$$\gamma C_v - C_v = R$$

$$(\gamma - 1)C_v = R$$

$$C_v = \frac{R}{\gamma - 1} = \frac{8.31}{1.4 - 1} = 20.78 \text{ J.mole}^{-1}\text{K}^{-1}$$

And,

$$C_p = \gamma C_v = 1.4 \times 20.78 = 29.09 \text{ J.mole}^{-1}\text{K}^{-1}$$

Second Law of Thermodynamics

Different Statements of The Law:

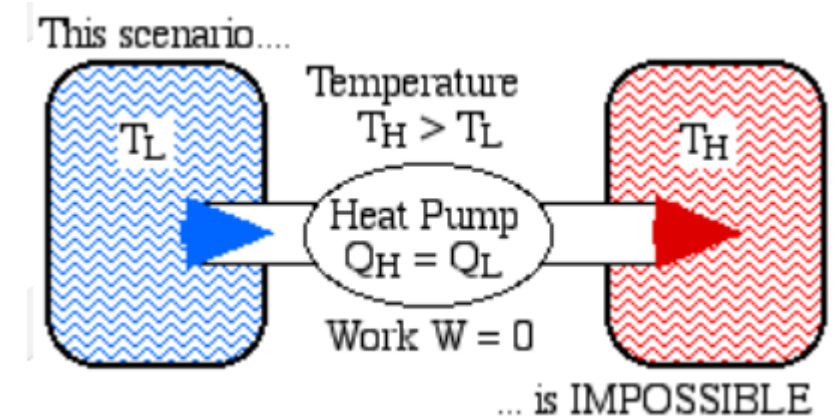
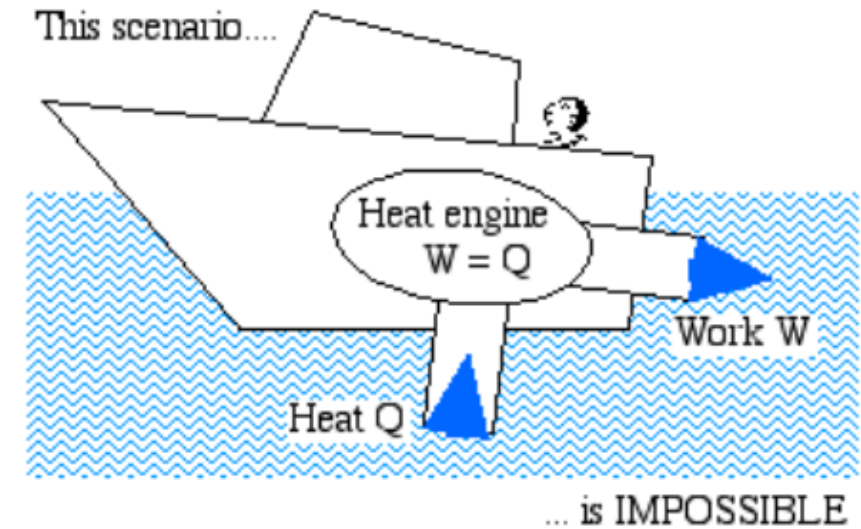
Kelvin- Plank Statement

It states that practically a reservoir never gives a positive net amount of work from the heat extracted from a thermal reservoir. So, basically, you cannot have a heat engine that operates between the two temperature levels and it has no heat rejection.

It is not possible to achieve a continuous supply of work from a body by cooling it to the temperature below the coldest of its surroundings.

Clausius Statement

It is impossible to construct a device operating in a cycle that can transfer heat from a colder body to warmer without consuming any work. Refrigerator works on Clausius's statement. In other words, unless the compressor is driven by an external source, the refrigerator won't be able to operate.



Entropy

Entropy is a measure of the randomness of the system or it is the measure of energy or chaos within an isolated system. Entropy is a State Function because it depends only on the initial and final thermodynamic states and not on the path followed.

It is denoted by 'S'. Mathematically, $S_2 - S_1 = \int_{Q_1}^{Q_2} \frac{dQ}{T}$. Here, dQ is the energy transferred as heat, T is the temperature of the system in Kelvins. Q_1 and Q_2 stands for initial heat and final heat of the system respectively.

Entropy change of a Perfect gas

Suppose that the initial temperature, pressure and volume of m kg of a gas be T_1 , P_1 and V_1 respectively. Let this gas be heated to final temperature, pressure, volume T_2 , P_2 , V_2 respectively.

Now, according to the 1st law of thermodynamics we can write,

$$dQ = dU + dW \dots \dots \dots (i)$$

Again, we can write,

$$dW = PdV$$

$$\text{And, } dU = nC_v dT$$

Putting these two values in Eqn. (i) we get,

$$dQ = nC_v dT + PdV \dots \dots \dots (ii)$$

From ideal gas law we know,

$$PV = nRT$$

$$\text{or, } P = \frac{nRT}{V}$$

Inserting this value in Eqn. (ii) we get,

$$dQ = nC_v dT + \frac{nRT}{V} dV$$
$$\Rightarrow \frac{dQ}{T} = nC_v \frac{dT}{T} + \frac{nRT}{TV} dV \quad [\text{dividing by } T]$$

Integrating,

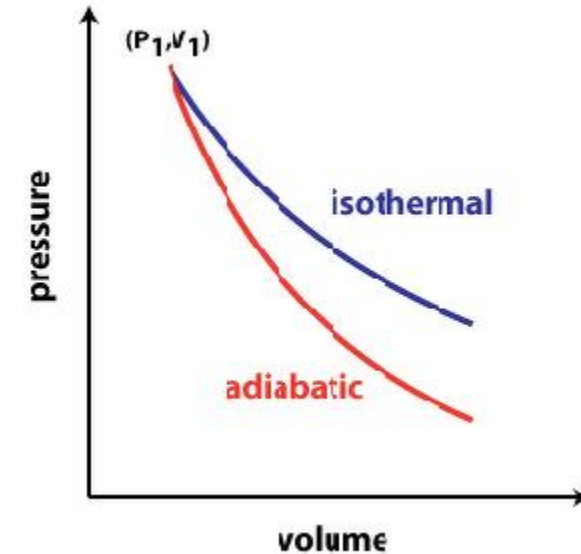
$$\Rightarrow \int_{Q_1}^{Q_2} \frac{dQ}{T} = \int_{T_1}^{T_2} nC_v \frac{dT}{T} + \int_{V_1}^{V_2} nR \frac{dV}{V}$$
$$\Delta S = S_2 - S_1 = nC_v \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1}$$
$$\Delta S = S_2 - S_1 = n \left(C_v \ln \frac{T_2}{T_1} + R \ln \frac{V_2}{V_1} \right)$$

This the required expression for change in entropy.

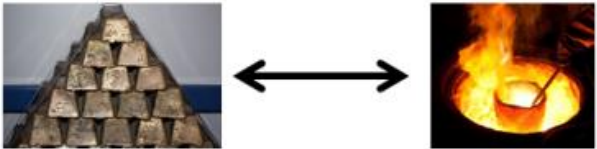
Problem: Calculate the increase in entropy of 2 gm of O₂ when its temperature is raised from 200 C to 1000 C and its volume is doubled. [C_v=25.5 Jmole⁻¹K⁻¹ and R=8.31Jmole⁻¹K⁻¹]

Difference Between Isothermal Process and Adiabatic Processes

Isothermal Process	Adiabatic Process
An isothermal process is a thermodynamic process in which there is no change in the temperature of the system. The temperature of the system remains constant throughout the thermodynamics process i.e. $\Delta T = 0$.	While an adiabatic process is the one in which there is no transfer of heat or mass in-between the system and the surrounding throughout the thermodynamic process. Hence, in an adiabatic system $\Delta Q = 0$.
From a thermal reservoir near the system, heat can be added or removed from the system, in order to keep the temperature constant.	By definition, there is no change in the heat in an adiabatic process and so there is no addition or subtraction of the heat.
Any transformation in such a process is slow.	Any transformation in such a process is fast.
Boyle's law holds good. i.e. $PV = \text{constant}$	Boyle's law doesn't hold good. i.e. $PV^\gamma = \text{constant}$



Differentiate between reversible and irreversible process.

Reversible process	Irreversible process	
1) It is a slow process which goes through various smaller stages which maintains equilibrium between the system and the surroundings.	(1) It is a comparatively fast process. Here, no equilibrium is maintained between the system and the surroundings.	REVERSIBLE PROCESS
2) Reversible processes can take place either in forward direction or in backward direction.	2) Irreversible processes can take place only in one direction.	
3) Reversible process proceeds in smaller steps because the driving force is small.	3) Irreversible process proceeds in larger steps in comparison with reversible process because there is definite driving force required for the progress of the process.	IRREVERSIBLE PROCESS
4) Work done in a reversible process is greater than the work done in an irreversible process.	4) Work done in an irreversible process is always lower than the work done in an irreversible process.	
(5) A reversible process can be brought back to the initial state without making any change in the surroundings	5) An irreversible process cannot be brought back to its initial state without making a change in the surroundings.	

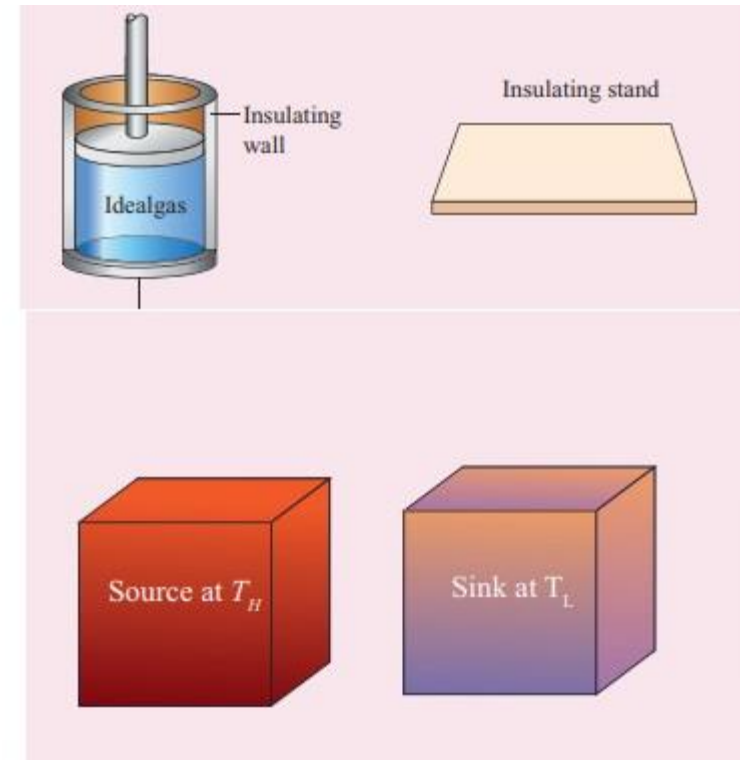
Carnot's Ideal Heat Engine

we have seen that the heat engine cannot have 100% efficiency. What is the maximum possible efficiency can a heat engine have?. In the year 1824 a young French engineer Sadi Carnot proved that a certain reversible engine operated in cycle between hot and cold reservoir can have maximum efficiency. This engine is called Carnot engine.

A reversible heat engine operating in a cycle between two temperatures in a particular way is called a Carnot Engine.

The carnot engine has four parts which are given below.

- i. **Source:** It is the source of heat maintained at constant high temperature T_H . Any amount of heat can be extracted from it, without changing its temperature.
- ii. **Sink:** It is a cold body maintained at a constant low temperature T_L . It can absorb any amount of heat.
- iii. **Insulating stand:** It is made of perfectly non-conducting material. Heat is not conducted through this stand.
- iv. **Working substance:** It is an ideal gas enclosed in a cylinder with perfectly non-conducting walls and perfectly conducting bottom. A non-conducting and frictionless piston is fitted in it.



Carnot's cycle

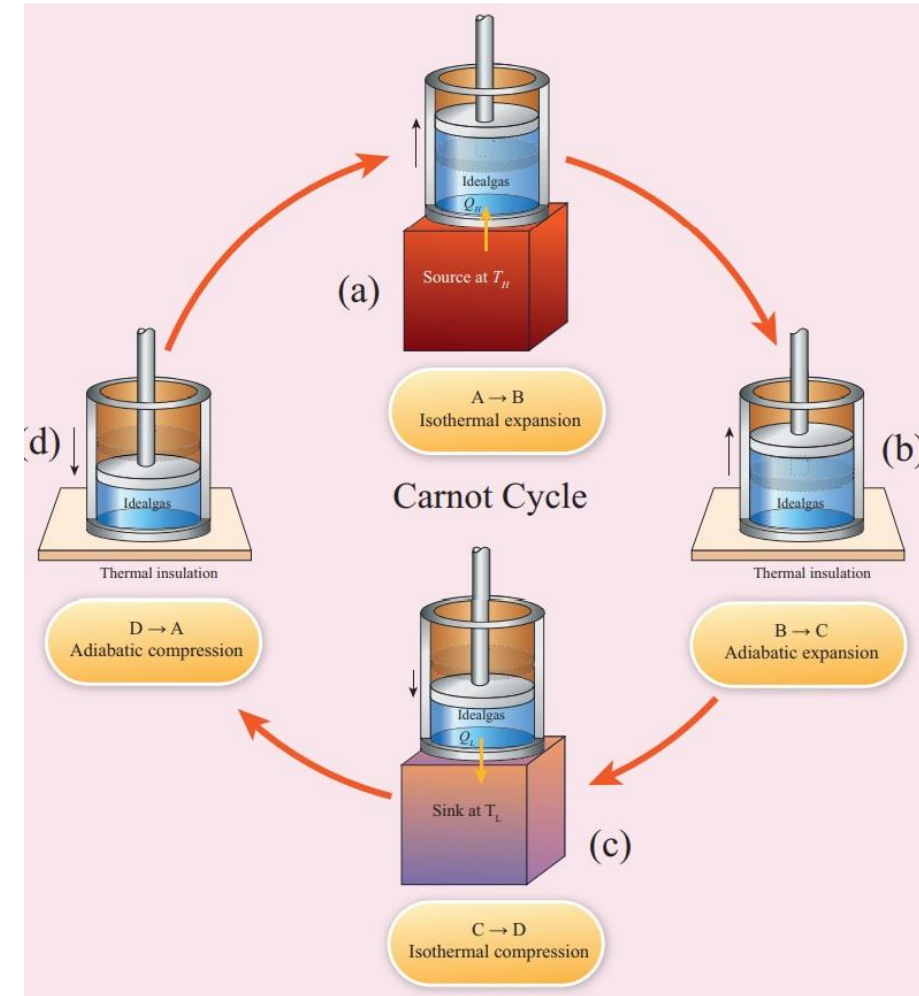
The working substance is subjected to four successive reversible processes forming what is called Carnot's cycle.

Let the initial pressure, volume of the working substance be P_1, V_1 .

Step A to B: Isothermal expansion from (P_1, V_1, T_H) to (P_2, V_2, T_H) :

The cylinder is placed on the source. The heat (Q_H) flows from source to the working substance (ideal gas) through the bottom of the cylinder. Since the process is isothermal, the internal energy of the working substance will not change. The input heat increases the volume of the gas. The piston is allowed to move out very slowly.

W_1 is the work done by the gas in expanding from volume V_1 to volume V_2 with a decrease of pressure from P_1 to P_2 . This is represented by the P-V diagram along the path AB.



Step B to C: Adiabatic expansion from (P_2, V_2, T_H) to (P_3, V_3, T_L)

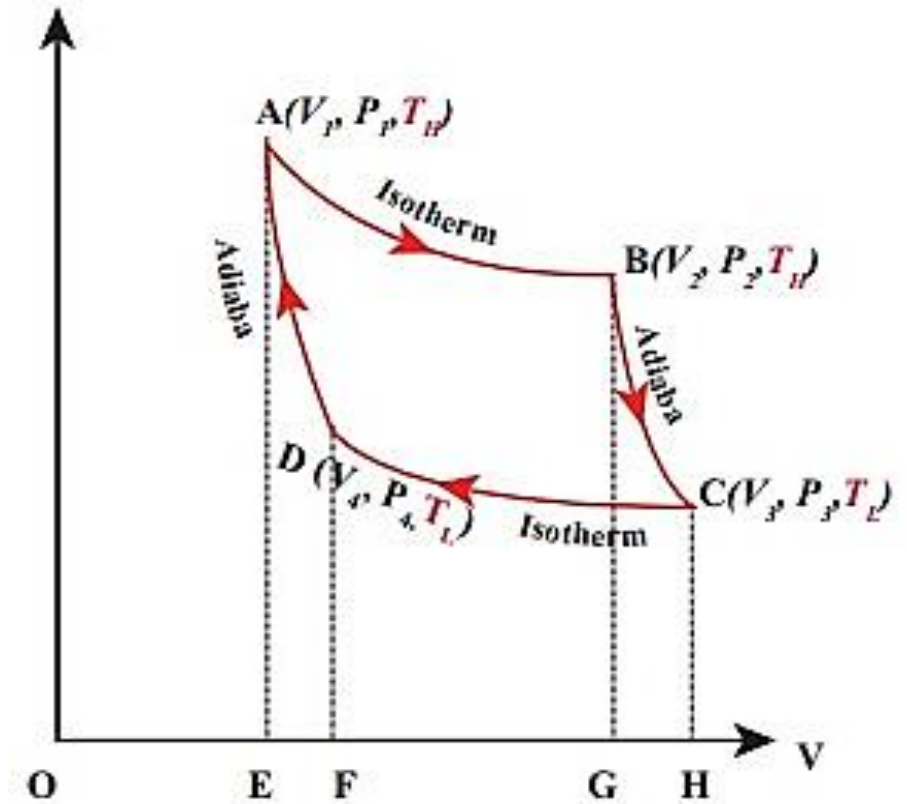
The cylinder is placed on the insulating stand and the piston is allowed to move out. As the gas expands adiabatically from volume V_2 to volume V_3 the pressure falls from P_2 to P_3 . The temperature falls to T_L . This adiabatic expansion is represented by curve BC in the P-V diagram. The work done by the gas in an adiabatic expansion is given by W_2 .

Step C to D: Isothermal compression from (P_3, V_3, T_L) to (P_4, V_4, T_L) .

The cylinder is placed on the sink and the gas is isothermally compressed until the pressure and volume become P_4 and V_4 respectively. This is represented by the curve CD in the PV diagram. Now the work done on the gas is W_3 . Work is done on the gas so that work done is negative

Step D to A: Adiabatic compression from (P_4, V_4, T_L) to (P_1, V_1, T_H) .

The cylinder is placed on the insulating stand again and the gas is compressed adiabatically till it attains the initial pressure P_1 , volume V_1 and temperature T_H . This is shown by the curve DA in the P-V diagram. In the adiabatic compression also work is done on the gas so it is negative



Let 'W' be the net work done by the working substance in one cycle

∴ W = Work done by the gas – work done on the gas

= W₁ + W₂ - W₃ - W₄ [since, W₂ = W₄]

= W₁ - W₃

Efficiency: The ratio of part of heat converted into work to the part of heat absorbed from the source is the efficiency of the engine.

$$\text{Efficiency, } \eta = \frac{\text{net work done by gas}}{\text{heat absorbed by gas}} = \frac{W}{Q_H} = \frac{Q_H - Q_L}{Q_H}$$

$$\text{or, } \eta = 1 - \frac{Q_L}{Q_H}$$

$$\text{Also we can write, } \eta = 1 - \frac{T_L}{T_H}$$

Third Law of Thermodynamics

- The third law of thermodynamics states that the entropy of a perfect crystal at a temperature of zero Kelvin (absolute zero) is equal to zero. This law was developed by the German chemist Walther Nernst between the years 1906 and 1912.
- At a temperature of zero Kelvin, the following phenomena can be observed in a closed system:
 - i. The system does not contain any heat.
 - ii. All the atoms and molecules in the system are at their lowest energy points.
 - iii. Therefore, a system at absolute zero has only one accessible microstate – its ground state. As per the third law of thermodynamics, the entropy of such a system is exactly zero.